Project Report — Mint (iPreneur) AI Pitch Deck Generator

Background, product, development history, technical overview, status and next steps

Infopace

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Repository guide

Mint turns a company website or a short questionnaire into an investor-grade deck. It was built between May and October 2026 by a single committer (57 commits), runs on Firebase Hosting and Cloud Run per the project log, and now needs hardening and a few unfinished features before wider launch.

Project at a glance

Item	Detail
Product	Mint (internal codename iPreneur), an AI pitch-deck generator
What it does	Crawls a company website (or takes a founder questionnaire), researches the market, writes every slide, audits the claims, renders a PowerPoint deck
Owner	Infopace Management Pvt. Ltd. (proprietary)
Repository	github.com/Infopace/Pitch-Deck; application in ipreneur/
Build period	20 May to 30 Oct 2026 (project log and git history)
Contributors	57 commits from a single committer
Users and plans	Founders; Free (₹0) and Pro (₹2,500/month via Razorpay)
Deck types	14 (investor, sales, product, partnership, internal, update, vision, general, elevator, demo day, go-to-market, business plan, master, appendix)
Themes	40 in four families
Stack	React 18 and TypeScript frontend; FastAPI and Python 3.11 backend; Google Gemini for AI; PostgreSQL; Playwright crawler; python-pptx
Hosting (per log)	Frontend on Firebase Hosting; backend on Cloud Run (asia-northeast1); data on Supabase
Size	About 12,800 lines of backend Python, 13,500 lines of frontend TypeScript, 559 passing backend tests
Status	Working end to end; needs security hardening, quota enforcement and a few unbuilt features (see Current status)

Background and objectives

Founders need a different deck for each room (investors, customers, partners, the board), and most tools produce one generic set of slides with figures nobody has checked. Mint aims to turn a company URL into a defensible deck in minutes, with each deck type written for its own audience.

Objectives, as stated in the repository

- **Speed:** from a URL to an investor-grade PPTX in minutes (the README cites about 2.5 minutes for the original pipeline).
- **Credibility:** claims grounded in web research and checked against the founder's own figures by a deterministic auditor, with sources captured.

- **Fit:** 14 deck types with their own slide sequences, stage-aware structure (pre-revenue founders are never asked to invent revenue), and Indian conventions (rupees, lakh and crore).
- **Control:** edit text in place, switch among 40 themes, export to PowerPoint.
- **Commercial model:** a Free plan and a Pro subscription billed through Razorpay.

Target users

Indian and global startup founders at idea, pre-seed, seed and growth stages, plus accelerators and venture studios (the dashboard was redesigned in an accelerator and venture-studio style, log #41 and #43).

Market positioning

The project's own research (log #56 and `reports/Pitch deck parameters for iPreneur.md`) found no competitor documenting claim-level source checking, stage and sector-aware structure, and India-native formats together. That is the stated open position.

Product

A founder pastes a URL (or fills a form), picks a deck type, and gets an editable, themed deck with a downloadable PPTX.

User journey

1. Land on the marketing page, preview themes live, register.
2. On the New Deck form, choose a deck type; the form shows only the fields that type uses and a preview of its slide order. Choose ₹ or $, upload a logo and up to five photos.
3. Watch progress (crawl, research, generate, render, save) on the workspace page.
4. Review the deck, edit any text in place, switch theme, and export to PowerPoint.
5. Upgrade to Pro through Razorpay when the Free limit matters.

Deck types

Group	Types (slides)
Raise capital	Investor (24, or 22 for idea and pre-seed), Elevator (5), Demo Day (10), Business Plan (25), Extended/Master (25)
Win customers	Sales (14), Product/Demo (14), Go-to-Market (22)
Align stakeholders	Partnership (15), Internal Decision Memo (10), Investor Update (8)
Explain the company	General Overview (20), Vision/Mission (8)
Evidence	Appendix (2 fixed bookends plus sections computed from the company profile)

Internal memos and Investor Updates do no web research, so nothing is invented.

Quality features

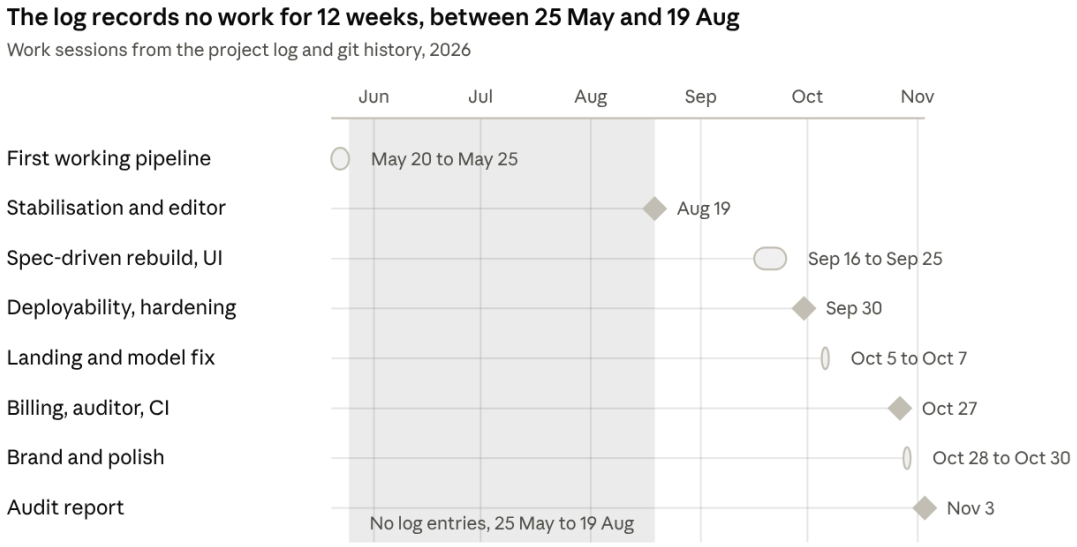
- A deterministic claims auditor checks each deck for invented revenue, inverted market funnels, percentages not totalling 100, unevidenced certifications and leftover placeholders.
- A company profile of 924 parameters records the source, method, confidence and date of each fact.
- 40 light and dark themes in four families; the landing page previews each theme with a different sample deck.
- Branded error, offline and generation-failure screens.

Plans

Plan	Price	Marketed limits
Free	₹0	3 decks a month, basic templates, PPTX export
Pro	₹2,500 a month	50 decks a month, premium templates, PDF export, priority AI
Enterprise	Custom	Unlimited decks, custom branding, API access, team collaboration

Note: the limits are advertised but not enforced in code, and PDF export and the Enterprise features are not built; the audit report covers this.

Development timeline



Diamonds are single-day sessions and bars are multi-day runs. The git history in this clone starts on 30 Sep 2026, so the earlier phases are dated from the project log, which may not capture every day of work.

Milestones and deliveries

The project log (61 entries) records the work from 20 May 2026. The git history in this clone begins on 30 Sep 2026, so earlier dates come from the log alone.

Date	Milestone	What was delivered
20 May	First working pipeline	Fixed a blank-workspace crash; ran generation in-process when Celery was not running; rebuilt the research agent with live web search, a 14-slide content agent, a six-layout PPTX renderer and a rich web preview (log #1 to #10)
21 May	Six-layer AI architecture	Added an analysis agent (investor score, SWOT, scenarios), a branding and website-intelligence layer, and a new progress step; end-to-end test with a fresh Gemini key (log #11 to #19)
25 May	Local auth and generation verified	Confirmed the sign-in and deck flow end to end (log #20)
19 Aug	Stabilisation and new flows	Fixed the “future attached to a different loop” crash and the empty-deck thinking-budget bug; added the no-website intake for idea-stage founders; wired Supabase Storage; quality upgrade to renderer and prompt; inline text editor (log #21 to #29)
8 Sep	Security and infrastructure triage	Removed live credentials from .env.example (history not scrubbed); diagnosed Supabase pause and DNS failures; decided to stay on Supabase Postgres (log #30, #31)
16 Sep	Specification-driven rebuild	Hero redesign; gap analysis against the 924-parameter spec; repaired silent render failures; built the master parameter model; added six deck types, reaching 13 (log #32 to #39)
18 to 25 Sep	Product UI	New sign-up page, executive dashboard redesigns, Apple-style typography, infographic themes, templates gallery, two-pane deck workspace, hourglass loader, retro 404 screen; fabricated dashboard metrics removed (log #40 to #55)
30 Sep	Deployability and hardening	Render and Cloudflare deployment files; rate limiting and optional Sentry; production config guards; orphaned-job reconciliation; truncated-deck detection (git history)
5 to 7 Oct	Landing redesign and model fix	3D deck showcase hero; Gemini 2.5 Flash retired so the model moved to gemini-3.8-flash; each background thread given its own database engine; research confidence and sources surfaced (git history)
27 Oct	Commercial layer	Razorpay billing; claims auditor; dynamic appendix deck; intake overhaul; rupee support; research report on real-world deck parameters; keyless frontend deploy to Firebase via Workload Identity Federation (log #56 to #61)
28 to 30 Oct	Brand and polish	New Mint logo and landing content; three-tier error pages; offline fallback; form persistence; business-plan deck fields; clickable contact links; PPTX export speed fix (git history)
3 Nov	Audit	Application audit report added to the repository

Technical overview

A React single-page app talks to a single FastAPI container; deck generation runs in a background thread inside that container and calls Gemini, the crawler, Postgres and object storage directly.

Stack

Layer	Technology
Frontend	React 18, TypeScript (strict), Vite, Tailwind, TanStack Query, Zustand, html-to-image and pptxgenjs for browser export

Layer	Technology
Backend	Python 3.11, FastAPI, async SQLAlchemy, Pydantic v2, SlowAPI rate limiting, bcrypt and JWT, Loguru, Sentry (optional)
AI	Google Gemini over REST, model gemini-3.8-flash per config, Google Search grounding, two API keys splitting quota
Crawler	Playwright (headless Chromium) with BeautifulSoup
Documents	python-pptx on the server (editable), pptxgenjs in the browser (image slides)
Data	PostgreSQL (Supabase), 6 tables, JSON columns for AI output; S3-compatible storage
Billing	Razorpay subscriptions, signed checkout and webhooks

How a deck is made

1. **Crawl** (if a website is given): breadth-first, priority-scored, default 18 pages, same domain only.
2. **Research:** Gemini with Google Search grounding finds founders, competitors and market facts; founder-supplied facts override web results. Skipped for decks that must not invent anything.
3. **Generate:** one structured call writes every section in the deck type's exact order, with retries and truncation detection.
4. **Validate and audit:** each section is checked against a typed schema; the claims auditor scores the deck and stores its findings.
5. **Render and store:** python-pptx renders the chosen theme; the file goes to object storage; the project is marked ready.

Design decisions worth knowing

- **One registry:** the 14 deck types and 65 section specs live in one Python module that drives the prompt, validation, renderer and the frontend form; a test keeps the TypeScript copy in step.
- **Per-section tolerance:** a malformed section is dropped, never the whole deck.
- **Provenance:** every profile value records source, method, confidence and date, with founder answers outranking crawled or researched ones.
- **Money:** one module formats rupee and dollar figures, with lakh and crore, and the auditor catches wrong-currency symbols.

Deployment (per repository and log)

The frontend deploys to Firebase Hosting from GitHub Actions on pushes to main, using Workload Identity Federation (no stored key). The backend runs on Cloud Run, built by a Cloud Build trigger that is not in the repo. Data is on Supabase. `render.yaml`, `wrangler.jsonc` and `vercel.json` are earlier targets and are not authoritative.

Current status

The core product works end to end; the gaps are commercial enforcement, security hardening and a few features that are advertised or half-built.

Area	State	Notes
Sign-up, sign-in, dashboard	Done	No password reset, email verification or account deletion
Website crawl, research, generation	Done	In production per log; runs in an in-process thread
14 deck types and intake forms	Done	Forms driven by the registry
40 themes and in-place editor	Done	Server PPTX renders later theme families in the default look
PPTX export	Done	Browser path gives image slides; server path gives editable text
Claims auditor	Partial	Runs and is stored; no screen shows the findings
Company profile (924 parameters)	Partial	Backend and API only; no frontend screen
Razorpay billing	Partial	Built; per log, production keys and plan were not yet set on 27 Oct
Plan limits (3 and 50 decks)	Not built	Advertised, never enforced
PDF export	Not built	Advertised for Pro
Enterprise features	Not built	Custom branding, API access, team collaboration
Celery workers	Legacy	Present, not on the live path

Metrics

Measure	Value
Commits	57 (single committer)
Backend Python	12,842 lines
Frontend TypeScript	13,471 lines
Backend tests	559 passing, 1 skipped, 0 failing (9 offline modules)
Frontend tests	None
Deck types, sections, parameters, themes	14, 65, 924, 40
API routes	30 versioned
Project log	1,374 lines, 61 entries

A separate audit report (`reports/Mint-iPreneur-Application-Audit-Report`) rates security, compliance and operations in detail.

Challenges and lessons

The project log records the root cause of each incident, which makes these the most reusable lessons.

Incident	Cause	Fix	Lesson
Every analysis crashed with “future attached to a different loop” (log #21)	A background thread used a database connection owned by the web server’s event loop	A separate no-pool engine per background loop	Keep async resources inside the loop that owns them

Incident	Cause	Fix	Lesson
Decks came back with no slides (log #22)	Gemini's thinking tokens consumed the output budget and truncated the JSON	Set the thinking budget explicitly; detect truncation instead of parsing it	Budget model output and check for truncation
Generation silently stopped after a Gemini model was retired (Oct)	gemini-2.5-flash closed to new users	Moved to gemini-3.8-flash	Probe model availability at start-up and alert
Dashboard spun forever in production	A new column existed in the model but not the database, and create_all never alters tables	Drift check and a migration script	Adopt real migrations
Jobs stuck on "generating" after a deploy	Generation runs in a thread that dies with the process	Start-up marks orphaned jobs failed	Move to a queue
Real credentials committed to .env.example (log #30)	Original secrets left in a comment block	Block deleted, not rotated or scrubbed from history	Rotate, then scrub, then guard with scanning
Pro price almost shown as ₹29 (log #59)	A currency symbol was swapped without changing the amount	Corrected to ₹2,500	Keep displayed prices tied to the billing plan
Whole-deck failure from one bad chart (commit 6b66650)	A market-chart crash aborted generation	Fixed the renderer; sections are validated independently	Isolate failures per section
PPTX export took about 21 s for 24 slides (commit 2c3a010)	Font CSS re-fetched for every slide	Compute font CSS once and reuse it	Measure before optimising
Frontend could not deploy with a service-account key	Organisation policy blocks key creation	Workload Identity Federation (log #61)	Prefer keyless CI

Open work and next steps

The first priority is closing four high-severity gaps before any wider launch; feature work follows.

Declared in the project log

- Rotate the credentials exposed in an earlier commit and scrub git history (log #30, #52).
- Profile UI and AI pre-fill; the selection engine and evidence layer (Phases 2, 4, 5, paused in log #52).
- Create the Razorpay plan and set production billing variables (log #59).
- Move generation onto the Celery worker so it survives restarts (comment in render.yaml and main.py).
- Server PPTX renders later theme families as the default theme (log #52).

Recommended priorities

Priority	Action	Why
1	Authenticate or delete the unauthenticated /pipeline/generate endpoint; add destination checks to the crawler; rotate keys	Cost abuse, server-side request forgery, exposed credentials

Priority	Action	Why
2	Enforce plan limits and meter usage	Advertised Free and Pro limits do nothing today
3	Add CI: backend tests, frontend build, dependency and secret scans	Nothing gates a merge to main
4	Show claims-audit findings in the editor	The main trust feature is invisible to users
5	Ship the profile screen	The 924-parameter engine has no user-facing surface
6	Queue-based generation, real migrations, hard-delete and account deletion	Reliability and privacy
7	Deliver or remove PDF export and Enterprise features	Marketed but not built
8	Refresh README and architecture docs	They describe an earlier Claude and Stripe design

Repository guide

Path	Contents
ipreneur/backend/app/	FastAPI app: api (routes), agents (master, audit, legacy), decks (registry, sections, prompt), profile (924 parameters), ppt (renderer, themes), services (crawler, storage, images), core, db, models
ipreneur/backend/test_*.py	11 test modules, 9 run offline
ipreneur/frontend/src/	11 pages, shared UI, workspace and deck-template components (TemplatedDeck, DeckWorkspace, exportDeck), API services, auth store
ipreneur/PROJECT_LOG.md	61-entry engineering log: problem, cause, fix, files
ipreneur/scripts/db/	init.sql and migrate.py (no Alembic)
ipreneur/docker-compose.yml	Development stack: Postgres, Redis, MinIO, backend, worker, beat, frontend
.github/workflows/	Frontend deploy to Firebase Hosting
render.yaml, wrangler.jsonc, vercel.json	Earlier deployment targets, not authoritative
reports/	Pitch-deck parameter research and the audit report (Markdown, Word, PDF)
research_notes/	Six notes on competitors, investor behaviour, India fundraising norms
Root documents	Bootstrap blueprint, Pitch Deck Parameters.docx (the 924-parameter spec), design archives, standalone demo HTML

Do not rely on README.md, ipreneur/README.md or docs/architecture/ARCHITECTURE.md; they describe an earlier design. Prefer the code and PROJECT_LOG.md.